

```

1 namespace CSharp;
2 class FunctionTest
3 {
4     public int fact(int n)
5     {
6         if (n == 0)
7             return 1;
8         else
9             return n * fact(n - 1);
10    }
11
12    public static int pow(int x ,int y)
13    {
14        if (y == 0)
15            return 1;
16        else
17            return x * pow(x, y - 1);
18    }
19    public int add(int x, int y)
20    {
21        Console.WriteLine("2 params");
22        return x + y;
23    }
24
25    public int add(int x = 1, int y = 2, int z = 3)
26    {
27        return x + y + z;
28    }
29    public float add(float x, float y)
30    {
31        return x + y;
32    }
33    public void max(int x, int y, out int mx)
34    {
35        if (x > y)
36            mx = x;
37        else
38            mx = y;
39    }
40    public void swap(ref int x, ref int y)
41    {
42        int temp = x;
43        x = y;
44        y = temp;
45        System.Console.WriteLine("inside swap x={0} Y={1}", x, y);
46    }
47    public static void FunMain()
48    {
49        FunctionTest ft = new FunctionTest();
50        Console.WriteLine("Factorial : {0}", ft.fact(5));
51        Console.WriteLine("Power : {0}", FunctionTest.pow(2, 3));
52        Console.WriteLine("Add int : {0}", ft.add(10, 20));
53        Console.WriteLine("Add int : {0}", ft.add(10, 20, 30));
54        Console.WriteLine("Add int : {0}", ft.add(10));
55        Console.WriteLine("Add int : {0}", ft.add());
56
57        Console.WriteLine("Add float : {0}", ft.add(10.5f, 20.5f));
58        int max;
59        ft.max(10, 20, out max);
60        Console.WriteLine("Max : {0}", max);
61        int a = 10, b = 20;
62        Console.WriteLine("Before swap a={0} b={1}", a, b);
63        ft.swap(ref a, ref b);
64        Console.WriteLine("After swap a={0} b={1}", a, b);
65    }
66 }
67
68
69
70
71
72

```